

# Eugenio ("Henny") Frias-Miranda

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## Research Interests

- Eugenio ("Henny") is a PhD candidate in Mechanical Engineering at Purdue University from San Juan, Puerto Rico. He works in Laura Blumenschein's Robust and Adaptive Design Lab. He is supported by the William E. and Florence E. Perry Teaching Fellowship, where he lectures a section of 109 students in Dynamics. His broader work centers on integrating soft robots into everyday life by leveraging compliant human-robot interaction, where human perception and decision-making are combined with soft robotic models to enable new capabilities and emergent behaviors. At present, he studies how wearable kinesthetic force feedback devices influence human motor adaptation, with an emphasis on how force magnitude and feedback design shape learning in full-body tasks. Previously, he has contributed to the development of a self-sensing obstacle-interacting vine robot and folded pneumatic artificial muscles. In his free time, Eugenio enjoys playing tennis, hiking, and learning about technology.

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## Education

Purdue University – West Lafayette, Indiana

- Doctor of Philosophy in Mechanical Engineering Aug 2022 – May 2027 | GPA: 3.90/4.00  
Advisor – Laura H. Blumenschein  
Dissertation – Leveraging Compliant Human Robot Interaction for Emerging Behaviors
- Bachelor of Science in Mechanical Engineering Aug 2018 – May 2022 | GPA: 3.50/4.00

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## Publications

### Submitted

- S1. **E. Frias-Miranda\***, A. de la Torre\*, D. Yu, and L. H. Blumenschein, "Soft Assumptions: Effect of Motion Scaling on Teleoperation of Soft Continuum Robots in Target-Reaching Tasks", Submitted.
- S2. **E. Frias-Miranda**, M. M. Matthews, M. R. Misterka, D. Chan, and L. H. Blumenschein, "Role of Kinesthetic Force Magnitude in Motor Learning After-Effects", In review.

### Peer-Reviewed Journal Articles

- J1. S. Wang, **E. Frias-Miranda**, and L. H. Blumenschein, "The Folded Pneumatic Artificial Muscle (foldPAM): Towards Programmability and Control via End Geometry," in IEEE Robotics and Automation Letters (RA-L), 2023.

### Peer-Reviewed Conference Papers

- C1. S. Wang, **E. Frias-Miranda**, A. Alvarez Valdivia, and L. H. Blumenschein, "Anisotropic Stiffness and Programmable Actuation for Soft Robots Enabled by an Inflated Rotational Joint," in IEEE International Conference on Robotics and Automation (ICRA), 2025.
- C2. **E. Frias-Miranda**, H. A. Nguyen, J. Hampton, T. Jones, B. Spotts, M. Cochran, D. Chan, and L. H. Blumenschein, "Adaptation and Training Effects from a Passive, Wearable Resistance Device During Exercise," in IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob), 2024.
- C3. **E. Frias-Miranda**, A. Srivastava, S. Wang, and L. H. Blumenschein, "Vine Robot Localization Via Collision", in IEEE International Conference on Intelligent Robots and Systems (IROS), 2023.

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## Presentations

- P1. **E. Frias-Miranda**, D. McTume, L. H. Blumenschein, "Toward a Modular Framework for Soft Growing Robots", in Society of Engineering Science (SES) Conference 2026. Poster Presentation.
- P2. **E. Frias-Miranda\***, A. de la Torre\*, D. Yu, and L. H. Blumenschein, "Effect Of Motion Scaling On A Soft Continuum Robot Target-Reaching Task, in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. Late Breaking Results.
- P3. **E. Frias-Miranda**, M. M. Matthews, M. R. Misterka, D. Chan, and L. H. Blumenschein, "Role of Kinesthetic Force Magnitude in Motor Learning After-Effects," in IEEE International Conference on Robotics and Automation (ICRA), 2025. Workshop Poster Presentation.
- P4. **E. Frias-Miranda**, H. A. Nguyen, J. Hampton, T. Jones, B. Spotts, M. Cochran, D. Chan, and L. H. Blumenschein, "Adaptation and Training Effects from a Passive, Wearable Resistance Device During Exercise," in IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob), 2024. Poster Presentation.
- P5. **E. Frias-Miranda**, H. A. Nguyen, J. Hampton, T. Jones, B. Spotts, M. Cochran, D. Chan, and L. H. Blumenschein, "A Wearable Resistance Device's Motor Learning Effects in Exercise" in Dynamic Walking Conference 2024. Poster Presentation.
- P6. **E. Frias-Miranda**, A. Srivastava, S Wang, and L. H. Blumenschein, "Vine Robot Localization Via Collision", in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023. Oral and Poster Presentation.
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## Honors and Awards

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|--------------------------------------------------------------------------------|----------|
| H1. Rising Stars in Mechanical Engineering   Stanford University               | Sep 2026 |
| H2. NextProf Nexus Workshop   Georgia Tech                                     | Aug 2026 |
| H3. William E. and Florence E. Perry Teaching Fellowship   Purdue University   | Jan 2026 |
| H4. Engineering Student Council EXPO Gold Scholarship   Purdue University      | Jan 2025 |
| H5. Dr. Anil Bajaj Outstanding Service Scholarship   Purdue University         | Dec 2024 |
| H6. Society of Hispanic Professional Engineers Scholarship   Purdue University | May 2023 |
| H7. Eagle Scout   Scouts of America                                            | May 2018 |

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## Teaching

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|-------------------------------------------------------------------------------------------|---------------------|
| <b>Lecturer</b>   Basic Mechanics II – Dynamics (ME 274)                                  | Jan 2026 – May 2026 |
| <b>Graduate Teaching Assistant</b>   Basic Mechanics I – Statics (ME 270)                 | Aug 2025 – Dec 2025 |
| <b>Undergraduate Teaching Assistant</b>   Systems, Measurements, and Controls II (ME 375) | Jan 2023 – May 2023 |

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## Professional Experience

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| <b>Mechatronics Fellow Intern</b>   GE Aerospace Research, Niskayuna, NY                                                                                                                                                                                                                                                                                                                               | Jun 2025 – Aug 2025 |
| <ul style="list-style-type: none"><li>Led concept design of a soft everting robot for jet engine inspection, contributing to a filed patent.</li><li>Deployed a soft everting robot in 50+ on-engine trials in Ohio, achieving a 75% deployment success rate.</li><li>Developed modular Rigidizable Guide Tube prototypes via thermal modeling and material testing, cutting iteration time.</li></ul> |                     |
| <b>Software Engineer Intern</b>   M365 Core, Microsoft, Redmond, WA                                                                                                                                                                                                                                                                                                                                    | May 2021 – Aug 2021 |
| <ul style="list-style-type: none"><li>Analyzed data gathered from a TDS machine and logged data analysis results in kusto using C#.</li><li>Created a processor which ingests a signal to obtain workplace data analytics.</li></ul>                                                                                                                                                                   |                     |
| <b>Program Manager Intern</b>   Project Agora, Microsoft Graph, Microsoft, Redmond, WA                                                                                                                                                                                                                                                                                                                 | May 2020 – Aug 2020 |
| <ul style="list-style-type: none"><li>Defined current problems, customers, and potential solutions within Microsoft Graph's API Review process.</li></ul>                                                                                                                                                                                                                                              |                     |

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## **Leadership & Service**

**President** | Mechanical Engineering Graduate Student Association

Aug 2022 – May 2026

- Organized major events for ME as direct link between ME students, faculty, and staff.
- Fostered communication skills by leading prospective student/new student orientation events.

**Volunteer** | 2025 IEEE International Conference on Robotics and Automation (ICRA 2025)

May 2025

**Volunteer** | 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2023)

Oct 2023

**Graduate Student Representative** | Society of Hispanic Professional Engineers

Aug 2019 – May 2023

- Build professional communication practices through networking and creating events with over 100 attendees.

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## **Technical Skills**

- **Software:** MATLAB, Python, C++, C#, C, Arduino, ROS, LabView, Motion Capture, EMGs, Unity
- **CAD:** SolidWorks, Siemens NX, CATIA, Creo Parametric, Abaqus, ANSYS

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## **Mentoring**

### **Undergraduate Students**

- U1. Javier Portilla Rivero, Purdue University, 2026–Present
- U2. Derek McTume, Purdue University, 2026–Present
- U3. Mackenzie M. Matthews, Purdue University, 2024–2026
- U4. Mackenzie M. Misterka, Purdue University, 2024–2025
- U5. Trenner A. Jones, Purdue University, 2023
- U6. Benjamin Spotts, Purdue University, 2023
- U7. Jeremy Hampton, Purdue University, 2023
- U8. Jeffrey Huang, Purdue University, 2023

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## **References**

**Dr. Laura H. Blumenschein** | Doctoral Advisor

Assistant Professor | School of Mechanical Engineering, Purdue University

1.765.494.5877 | lhblumen@purdue.edu

**Dr. Deva Chan** | Committee Member & Collaborator

Associate Professor | Weldon School of Biomedical Engineering, Purdue University

Courtesy Appointment | School of Mechanical Engineering, Purdue University

1.765.496.1768 | chand@purdue.edu

**Dr. Denny Yu** | Collaborator

Cubitt Rising Star Professor | Edwardson School of Industrial Engineering, Purdue University

Courtesy Appointment | School of Health Sciences, Purdue University

1.765.494.7346 | denny@purdue.edu

**Dr. Charles M. ("Chuck") Krousgrill** | Lecturing Colleague & Mentor

Tebbe Family 150th Anniversary Professor | School of Mechanical Engineering, Purdue University

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